DSC 40A - Homework 6

due Friday, July 24th at 11:59PM

Homeworks are due to Gradescope by 11:59PM on the due date.

You may request an extension for this assignment by submitting the form.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using LaTeXor any other software)**. Staff will not grade typed submissions.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

Notes:

- For full credit, make sure to **assign pages to questions** when you upload your submission to Gradescope.
- Throughout this homework, it's fine to leave answers unsimplified, in terms of factorials, exponents, the permutation formula $P(n, k)$, and the binomial coefficient $\binom{n}{k}$.

Problem 1. Reflection and Feedback Form

Make sure to fill out this Reflection and Feedback Form, linked here for two points on this homework! This form is primarily for your benefit; research shows that reflecting and summarizing knowledge helps you understand and remember it.

Problem 2. Baseball

a) Suppose that your baseball team has 12 players but you only have 8 matching team hats. How many ways are there for you to select who gets a hat and who doesn't?

b) Suppose that your baseball team has 12 players and you have 16 jerseys, numbered 1 through 16. You will give out one jersey to each player, and four will be leftover. How many ways are there for you to assign jerseys to players?

c) As before, your baseball team has 12 players and you have 16 jerseys, numbered 1 through 16. You will give out one jersey to each player, and four will be leftover. How many ways are there for you to assign jerseys to players if you must give out jersey number 7?

d) Suppose that at the batting cage, where a machine pitches the ball, each time you swing the bat, your probability of hitting the ball is 3/5. If you swing 9 times, what is the probability that you hit the ball exactly 7 times?

e) Suppose that when a pitcher is throwing the ball, your probability of hitting the ball depends on the kind of pitch. Your probability of hitting the ball is

- $\frac{1}{2}$ for a fastball,
- $\frac{1}{3}$ for a breaking ball, and
- $\frac{1}{4}$ for a changeup

Suppose that at practice, a pitcher throws you three fastballs, three breaking balls, and three changeups. What is the probability that you miss one breaking ball and one changeup, but hit all 7 other balls?

Problem 3. Cavocado

The tutors in DSC 40A also like eating at Cava, a Mediterranean restaurant off-campus. They also like eating avocados.

In this problem, we will consider various permutations of the string **CAVOCADO**. Unless specified, a permutation of CAVOCADO will be of the same length as CAVOCADO, i.e. also of length 8.

a) Prove that the number of permutations of CAVOCADO is:

$$\frac{8!}{2!2!2!}$$

Note: This is really meant to be the problem "How many permutations of CAVOCADO are there?" but we've framed it as a "proof" so that you know what the answer is supposed to be.

b) How many permutations of CAVOCADO start with a C?

c) How many permutations of CAVOCADO contain the letters CAD next to each other, in that order?

d) How many permutations of CAVOCADO contain the letters CAD next to each other, in any order?

e) How many permutations of CAVOCADO contain all of the vowels **before** all of the consonants? (A and O are vowels, and C, D, and V are consonants.)

f) Prove that the number of permutations of CAVOCADO of **length 4** is 354.

Hint: Break the problem into three cases.

Problem 4. Prime Factorization

We've recorded a hint/walkthrough video for this problem — we highly encourage you to watch it: https://youtu.be/Jx1GrwQ44HU

Recall, a **prime number** is a natural number greater than 1 that cannot be written as a product of smaller natural numbers. For instance, 2 and 3 are prime, while 6 and 24 are not (because $6 = 2 \cdot 3$ and $24 = 2 \cdot 12 = 3 \cdot 8 = ...$). Natural numbers that are not prime (and not equal to 1) are called **composite**.

The **Fundamental Theorem of Arithmetic** states that every natural number greater than 1 can be written as the product of prime factors, and that every natural number has a unique representation in this form. (If a is a factor of b, then $\frac{b}{a}$ is a whole number.)

For example, 12 can be written as $2^2 \cdot 3$, and 1200 can be written as $2^4 \cdot 3 \cdot 5^2$.

To compute the prime factorization of a number, one strategy is to repeatedly divide by the smallest prime factor that divides that number. For instance, here's the process of prime factoring 1200 ("pf" stands for "prime factor"):

$$\begin{aligned}
1200 &= 1200 \\
&= 2 \cdot 600 \quad \text{(2 is the smallest pf of 1200)} \\
&= 2 \cdot 2 \cdot 300 \quad \text{(2 is the smallest pf of 600)} \\
&= 2 \cdot 2 \cdot 2 \cdot 150 \quad \text{(2 is the smallest pf of 300)} \\
&= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 75 \quad \text{(2 is the smallest pf of 150)} \\
&= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 25 \quad \text{(3 is the smallest pf of 75)} \\
&= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 5 \cdot 5 \quad \text{(5 is the smallest pf of 25)} \\
&= 2^4 \cdot 3 \cdot 5^2
\end{aligned}$$

A handy shortcut is that if the number we are prime factoring ends in a 0, we know that is divisible by 10 and hence we can pull out a factor of $2 \cdot 5$.

You may be thinking, **what does this have anything to do with combinatorics?** Good question — we're getting there.

Suppose we now want to determine the number of factors that 1200 has. One rather painful way to do it would be to enumerate them manually — 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, etc. Of course, there's a better way.

Each factor of 1200 will be made up of some number of 2s, some number of 3s, and some number of 5s, all multiplied together. If you don't believe this, write out a few factors of 1200, and you will see that each of them is a product of 2s, 3s, and 5s. For example:

$$\begin{aligned}
24 &= 2^3 \cdot 3 \\
30 &= 2 \cdot 3 \cdot 5 \\
120 &= 2^5 \cdot 5
\end{aligned}$$

There are 5 options for the number of 2s a factor could have: 0, 1, 2, 3 or 4 (meaning a factor of 1200 could either have a factor of 2^0, or 2^1, or 2^2, or 2^3, or 2^4). Similarly, there are 2 options for the number of 3s a factor could have (either 0 or 1) and 3 options for the number of 5s (0, 1, or 2).

In other words, each factor of 1200 will look like

$$2^a \cdot 3^b \cdot 5^c$$

where

$$0 \le a \le 4 \qquad 0 \le b \le 1 \qquad 0 \le c \le 2$$

Since we're making three successive choices, we take the product of the number of choices at each step, yielding

$$\text{number of factors of } 1200 = (4+1)(1+1)(2+1) = 30$$

Note that this counts both 1 ($2^0 \cdot 3^0 \cdot 5^0$) and 1200 ($2^4 \cdot 3^1 \cdot 5^2$) as factors of 1200.

In this problem, you *must* use this approach to find the number of factors, and you must show your work.

a) Let's try out a few examples.

1. Compute the prime factorization of 2500, and use it to determine the number of factors of 2500.
2. Compute the prime factorization of 7260, and use it to determine the number of factors of 7260.

In both subparts, manually listing out all factors will not receive credit; you must use the counting technique outlined in the question.

b) How many factors of 2500 are multiples of 50?

Hint: Think about how this condition restricts the number of options for each prime factor.

c) How many factors of 7260 are multiples of 2 but not multiples of 121?

Again, think about the possibilities for the exponents on each of the prime factors.

Problem 5. Hockey-stick identity

You've learned in lectures that:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

which means the number of combinations of picking k elements out of n unique elements.

a) Prove Pascal's rule:

$$\binom{n+1}{k+1} = \binom{n}{k+1} + \binom{n}{k}$$

You may assume $n > k$. Then based on the meaning of $\binom{n}{k}$, give intuitive explanation for this rule.

In general, there is no restriction on the relative sizes of n and k. If $n < k$, the value of the binomial coefficient is zero and the identity remains valid.

b) Use Pascal's rule, show Hockey-stick identity:

$$\binom{n+1}{k+1} = \binom{n}{k} + \binom{n-1}{k} + \binom{n-2}{k} + \ldots + \binom{k}{k}$$

The Hockey-stick name comes from the visualized pattern in Pascal's triangle, in which each element's value equals the sum of its left-upper and right-upper neighbors:

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

$$\binom{4}{2} = \binom{3}{1} + \binom{2}{1} + \binom{1}{1}$$